

```
In [1]: import numpy as np # Linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
```

```
In [4]: import pandas as pd
data = pd.read_csv('spamm.csv', encoding="latin1")
# or
# data = pd.read_csv(r"C:\Users\Admin\Desktop\spamm.csv", encoding="ISO-8
data.head()
```

```
Out[4]:
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

```
In [5]: data = pd.read_csv('spamm.csv')
data
```

```
Out[5]:
```

	Category	Message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...
...	...	...
5567	spam	This is the 2nd time we have tried 2 contact u...
5568	ham	Will ü b going to esplanade fr home?
5569	ham	Pity, * was in mood for that. So...any other s...
5570	ham	The guy did some bitching but I acted like i'd...
5571	ham	Rofl. Its true to its name

5572 rows × 2 columns

```
In [6]: data.columns
```

```
Out[6]: Index(['Category', 'Message'], dtype='object')
```

```
In [7]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Category    5572 non-null   object
 1   Message     5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB
```

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In [8]: data.isna().sum()
```

```
Out[8]: Category    0
        Message     0
        dtype: int64
```

```
In [9]: data['Spam'] = data['Category'].apply(lambda x:1 if x == 'spam' else 0)
        data.head(5)
```

```
Out[9]:
```

	Category	Message	Spam
0	ham	Go until jurong point, crazy.. Available only ...	0
1	ham	Ok lar... Joking wif u oni...	0
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1
3	ham	U dun say so early hor... U c already then say...	0
4	ham	Nah I don't think he goes to usf, he lives aro...	0

```
In [10]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    data.Message,    # Features (input)
    data.Spam,      # Labels (target)
    test_size=0.2,  # 20% of the data for testing
    random_state=42 # for reproducibility
)
```

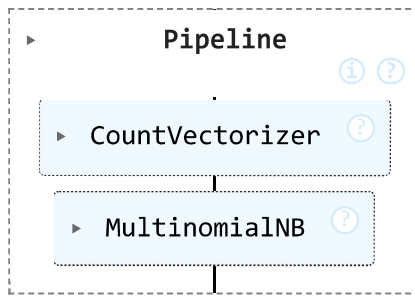
```
In [11]: #CounterVectorizer Convert the text into matrices
from sklearn.feature_extraction.text import CountVectorizer
```

```
In [12]: from sklearn.naive_bayes import MultinomialNB
```

```
In [22]: from sklearn.pipeline import Pipeline
clf=Pipeline([
    ('vectorizer',CountVectorizer()),
    ('nb',MultinomialNB())
])
```

```
In [20]: clf.fit(X_train,y_train)
```

Out[20]:



```
In [16]: emails=[  
    'Sounds great! Are you home now?',  
    'Will u meet ur dream partner soon? Is ur career off 2 a flyng start?'  
]
```

```
In [17]: clf.predict(emails)
```

```
Out[17]: array([0, 0])
```

```
In [18]: clf.score(X_test,y_test)
```

```
Out[18]: 0.9919282511210762
```

```
In [ ]:
```